Camilo Andres Duarte

candresdu@gmail.com +1 301 674-4987 linkedin.com/in/camilo-06bb03ab

SUMMARY

I am seeking a role as a Lead Robotics Software Engineer focused on autonomy, motion planning, or perception for autonomous systems. I have over 6 years of industry experience, most recently owning the end-to-end autonomy stack (perception, motion planning, controls, grasp planning, and embedded firmware) for a proprietary, custom-designed parcel handling robot deployed in customer pilots, with prior work deploying algorithms for off-the-shelf industrial arms (UR10e) and UAVs. My master's degree was focused on deep learning for motion planning of systems with highly nonlinear dynamics and non-convex cost functions.

PROFESSIONAL EXPERIENCE

CARGO ROBOTICS (Senior Robotics Software Engineer, Employee #2/Founding Team) | Menlo Park, CA | March 2025 – Present

- **Custom Autonomous Package-Loading Robot — End-to-End Autonomy Stack**
  - Joined as one of two software engineers and co-architected the autonomy and embedded stack for a ground-up, proprietary robotic platform, progressing the custom hardware from initial prototype to road-test and commercial deployment readiness.
  - Drove autonomy development from early demonstrations through regional customer evaluations, securing the company's first signed paid pilot with a major global carrier and expanding commercial capabilities to support international partner milestones.
  - Owned deployment and operational tooling, including automated sensor calibration frameworks, hardware diagnostics, evaluation harnesses, and fleet telemetry pipelines to ensure reliability in multi-shift customer environments.
- **Perception, Planning & Controls**
  - Co-designed planning and control frameworks for an in-house actuated end-effector (rotation and expansion), coordinating yaw alignment, kinematics, and footprint-aware grasp planning to maximize grasp accuracy across shifted and deformable parcels.
  - Developed a multi-stage trajectory collision checking and path validation framework within MoveIt, verifying clearances along the full motion profile to eliminate kinematic failure modes and prevent collisions in dense environments.
  - Overhauled near-field 3D perception with range-gated depth filtering and reprojection quality gating to eliminate grasp pose errors; engineered visual and spatial capture pipelines supplying structured inputs to downstream VLM decision interfaces for anomaly handling and parcel re-identification.
  - Architected a configuration-driven robot-model generation pipeline (URDF/SRDF), establishing the kinematic model and digital twin from scratch for the custom platform across hardware iterations.
- **Embedded Firmware, Deployment & Reliability**
  - Architected distributed embedded firmware for the actuated end-effector across multi-MCU boards over CAN, implementing closed-loop motion control, homing with lash compensation, and multi-zone vacuum telemetry for reliable parcel retention.
  - Transitioned the custom gantry drivetrain to closed-loop BLDC motor control over CAN with absolute encoder feedback and fail-safe braking, unifying hardware control layers across robot generations.
  - Engineered robust error-handling and recovery frameworks, enabling safe, in-place fault and e-stop recovery without system reboots, backed by CAN bus watchdogs and real-time hardware health monitoring.
  - Maintained containerized deployment and orchestration pipelines (Docker, ROS 2, Jetson Orin AGX) that standardized software across hardware revisions and reduced bare-metal bring-up to under 2 hours.

LOCKHEED MARTIN (AI/Robotics Research Engineer) | Palo Alto, CA | September 2021 – March 2024

- **Robotic Assisted Agility**
  - Realized an estimated $0.4M per year savings by leading the deployment of autonomy modules for wire harness assembly using robotic arms, reducing labor costs.
  - Designed a set of motion constraint functions for a UR10e robotic arm to simplify motion planning, enhancing safety and repeatability in dynamic manufacturing environments.
  - Innovated automated eye-in-hand camera calibration and a robust object pose-estimation framework with fiducial markers and a Kalman-filter sensor-fusion solution in C++, enabling millimeter-level accuracy for object picking.
  - Optimized Machine Learning models in the autonomy stack using TensorRT, improving inference runtime by 10x and enabling real-time processing.
- **Firefighting Intelligence as a Service** [Link]
  - Led a research team implementing and evaluating neural network architectures in PyTorch for wildfire behavior prediction, applying insights to real-world fire data in California and Colorado.

GEORGIA TECH RESEARCH INSTITUTE (Graduate Research Assistant) | Atlanta, GA | January 2020 – June 2021

- **AlphaDogfight DARPA Challenge**
  - Implemented a trajectory-tracking controller for a simulated F-16 (JSBSim) supporting AI-driven within-visual-range air-combat maneuvering.
  - Carried out system identification experiments to estimate aerodynamic force and moment coefficients at different flight conditions.
- **Combined Proportional Navigation for Object Tracking with Partial State Observation**
  - Engineered, tested and deployed various vision-based Proportional Navigation Guidance algorithms for UAV target tracking and pursuit missions of adversarial targets.

NASA LANGLEY RESEARCH CENTER (Engineering Intern) | Hampton, VA | January 2018 – May 2018

- **Augmenting Adaptive Control for the Space Launch System** [Paper]
  - Built a quadcopter testbed with the purpose of replicating launch vehicle fuel-sloshing dynamics, ran system identification studies, and implemented an Augmented Adaptive Controller for attitude control with an LQR for waypoint tracking.

RESEARCH EXPERIENCE

GEORGIA INSTITUTE OF TECHNOLOGY (Graduate Researcher) | Atlanta, GA | May 2019 – May 2021

- **Deep L1 Stochastic Optimal Control Policies for Planetary Soft-landing** [Paper]
  - Advanced non-convex optimal control for minimum-fuel powered-descent guidance, co-developing and implementing a deep-learning-based stochastic optimization algorithm in PyTorch; published in the Journal of Guidance, Control, and Dynamics.
- **Risk-Sensitive Gradient-Based Adaptive Stochastic Search** [Paper]
  - Developed a motion-planning algorithm minimizing Conditional Value-at-Risk for systems with nonlinear stochastic dynamics, demonstrating its efficacy on an inverted pendulum, a cartpole, and a quadcopter.
- **Alphapilot AI Drone Innovation Challenge**
  - Developed a Model Predictive Path Integral controller for autonomous drone racing, covering simulation testing, C++/CUDA cost-map generation, cost-function design, and Extended Kalman Filter state estimation.

EDUCATION

GEORGIA INSTITUTE OF TECHNOLOGY

- **M.S. in Aerospace Engineering** — **GPA:** 3.83/4.00 — **Aug 2019 - Jun 2021**
  - **Focus:** Robotics and Autonomous Systems
- **B.S. in Aerospace Engineering** — **GPA:** 3.87/4.00 — **Jan 2015 - Jun 2019**

PUBLICATIONS AND MANUSCRIPTS

1. M. Pereira, **C. Duarte**, I. Exarchos, and E. Theodorou. "Deep L1 Stochastic Optimal Control Policies for Planetary Soft-Landing" *Journal of Guidance, Control, and Dynamics*. https://arc.aiaa.org/doi/10.2514/1.G007132
2. Z. Wang, K. Lee, O. So, **C. Duarte**, and E. Theodorou. "Adaptive CvaR Optimization for Dynamical Systems with Path Space Stochastic Search," 2020. https://arxiv.org/pdf/2009.01090
3. J. Pei, A. Puetz, **C. Duarte**, and L. Miller. "Suppression of Nonlinear Rotary Slosh Dynamics using the SLS Adaptive Augmenting Control System Demonstration on a Quadcopter Testbed," *AIAA SciTech*, 2019. https://doi.org/10.2514/6.2019-0114

SKILLS & ABILITIES

Coding Languages and Development Platforms

- Python, C++, MATLAB, CUDA, Bash; Linux (Ubuntu, Red Hat), Docker (multi-arch), Git/GHCR, Jetson Orin AGX, ESP32/Arduino, CAN/CANopen.

Frameworks, Simulators and Libraries

- ROS/ROS 2, MoveIt; PyTorch, TensorFlow, TensorRT, OpenCV, Open3D, Vision-Language Models (VLMs); NVIDIA cuRobo (nvBlox/cuMotion), Kubeflow, Determined AI, KServe, GDAL/GeoPandas, JSBSim.